CS23532 – COMPUTER NETWORKS
EX 3: Cisco Packet Tracer

AIM :

To study the installation and user interface of the Cisco Packet Tracer simulation tool, and to design a simple network with a hub and a switch and analyse the behaviour of these network devices.

PROCEDURE :

Introduction

A simulator simulates network devices and their environment. Packet Tracer is a network design, simulation and modelling tool.

- It allows you to model complex systems without the need for dedicated equipment.
- It helps you practise network configuration and troubleshooting skills on a computer or an Android or iOS mobile device.
- It is available for both Linux and Windows desktop environments.
- Protocols in Packet Tracer are coded to work and behave in the same way as they would on real hardware.

Installing Packet Tracer

Go to https://www.netacad.com, log in with your Cisco Networking Academy credentials, click on the Packet Tracer graphic and download the package for your operating system.

Windows: Open the setup file (Packettracer_Setup6.0.1.exe) to begin the setup wizard, accept the license agreement, choose a location and start the installation.

Linux: Download the file for Ubuntu/Debian or for Fedora/Redhat/CentOS, grant executable permission using chmod and execute it to begin the installation.

```
chmod +x PacketTracer601_i386_installer-rpm.bin
./PacketTracer601_i386_installer-rpm.bin
```

User interface overview

1. Menu bar – A common menu used to open, save, print, change preferences, and so on.

2. Main toolbar – Shortcut icons to commonly used menu options such as open, save, zoom, undo and redo; on the right-hand side is an icon for entering network information for the current network.

3. Logical/Physical workspace tabs – Used to toggle between the Logical and Physical work areas.

4. Workspace – The area where topologies are created and simulations are displayed.

5. Common tools bar – Controls for manipulating topologies, such as select, move layout, place note, delete, inspect, resize shape and add simple/complex PDU.

6. Real-time/Simulation tabs – Used to toggle between the real-time and simulation modes. Buttons are also provided to control the time and to capture the packets.

7. Network component box – Contains all the network and end devices available in Packet Tracer. It is divided into the device-type selection box (7a), which contains the device categories, and the device-specific selection box (7b), which displays the device models of the selected category.

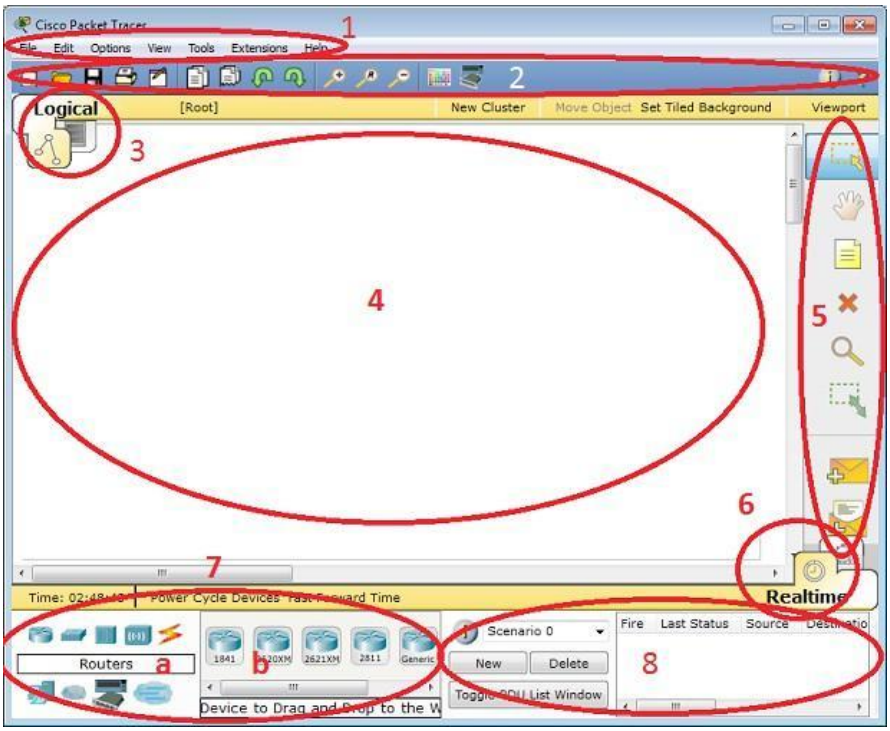
8. User-created packet box – Users can create customised packets to test their topology from this area, and the results are displayed as a list.

Analyse the behaviour of network devices

1. From the network component box, drag and drop 4 generic PCs and one hub, and 4 generic PCs and one switch.
2. Click on Connections and select the Copper Straight-Through cable. Connect each PC to the hub; the link LED glows green when the link is up. Similarly, connect the 4 PCs to the switch.
3. Click on a PC, go to the Desktop tab, click on IP Configuration and enter an IP address and subnet mask. The default gateway and DNS server are not needed in this network.
4. Click on the PDU (message icon) in the common tools bar and drag and drop it on the source PC and then on the destination PC connected to the hub.
5. Observe the flow of the PDU from the source PC to the destination PC using the Realtime mode of the simulation.
6. Repeat the same steps for the PCs connected to the switch.
7. Observe how the hub and the switch forward the PDU and note the behaviour of each device.

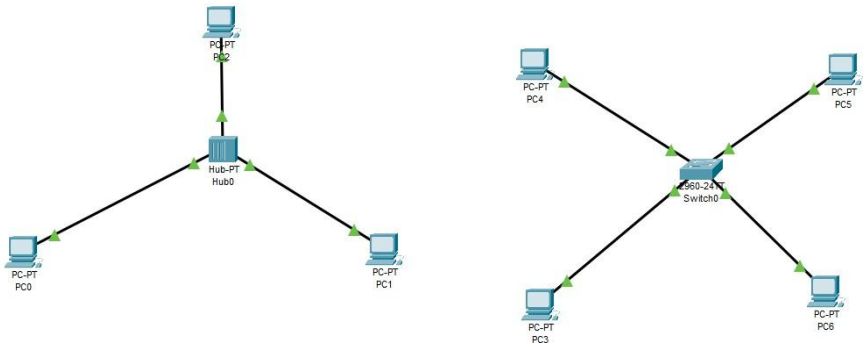
OUTPUT:

Packet Tracer user interface

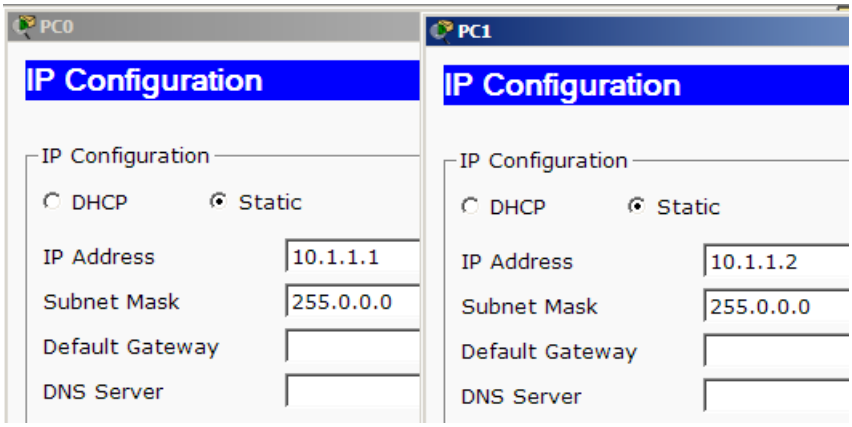


Network topology with a hub and a switch (Cisco Packet Tracer)





IP configuration of the PCs



RESULT:

Thus, the Cisco Packet Tracer environment was studied, and a simple network using a hub and a switch was designed and simulated successfully.